Day 61 - AI Monitoring & Observability Plan

Essential design, metrics, alerts, dashboard, and decision-audit controls

Objective

Track pipeline performance, detect failures, compare AI outputs with reviewed outcomes, and preserve an auditable history of decisions.

Current project state

- **Logging:** `utils/logger.py` writes plain-text application messages to `logs/app.log`.

- **Audit records:** `AuditLogger` writes score and decision JSONL files containing candidate IDs and decision details, but does not enforce access control, encryption, retention, or integrity checks.

- **Missing infrastructure:** no HTTP API, monitoring backend, alert delivery service, or dashboard UI is present. Existing benchmark results are local service measurements.

- **Quality-data gap:** completed human-review scores and adjudicated decisions are not stored as structured feedback. AI accuracy cannot yet be measured reliably.

Logging and key metrics

Use one JSON event per line with schema version, UTC timestamp, event ID/type, level, component, request/run ID, stage, status, duration, and safe error code. API events (when an API exists) add method, route template, and status code. Never log request bodies, tokens, resumes, or transcripts.

- **Latency:** end-to-end and per-stage p50/p95; keep model loading and warm-up separate.

- **Reliability:** terminal failures / accepted runs; track retries and stage errors separately.

- **AI quality:** matched, adjudicated human reviews only. Show score mean absolute difference, decision agreement, advance recall/precision, false-rejection rate, review coverage, and sample count. With no labels, show “not measurable.”

Provisional pilot alerts

- Critical: model readiness fails twice in a row, or a required audit write fails.

- Terminal failure rate above 5% in 5 minutes with 20+ runs: warning; above 10%: critical.

- End-to-end p95 above 1 second for 10 minutes with 100+ runs: warning; above 2 seconds for 5 minutes: critical.

- Retry rate above 10% / 25%, or invalid-output rate above 1% / 5%, in 15 minutes with 20+ events: warning / critical.

- With 50+ matched human reviews in 28 days, score difference above 10 points or advance recall below 90% prompts review; never auto-change candidate outcomes.

Thresholds are provisional. Day 60’s highest local p95 was about 788 ms at concurrency 4 in a small, warmed single-process run without HTTP or a load balancer. Recalibrate against representative deployed traffic before paging.

Dashboard, audit controls, and rollout

Dashboard sections

- **Pipeline health:** accepted/completed runs, completion and failure rates, stage funnel, p50/p95, top errors, model readiness, and active alerts.

- **Interview outcomes:** HR and technical stages separately. Completion = completed / scheduled; advancement = human-approved advances / completed interviews with a final human decision. Show pending, no-show, cancelled, and withdrawn separately.

- **AI quality:** matched AI-human metrics by role, stage, model version, review coverage, and sample count. Keep fairness-group analysis in a separately governed view.

- **Alerts and audit:** show severity, owner, and runbook. Link to individual audit records only through a protected service for authorized reviewers.

Decision audit requirements

Record score generation, decisions, completed human reviews, overrides, audit access, and retention actions as versioned, append-only events. Include event/time, opaque candidate reference, request/run ID, trusted actor/service, role/stage, model and policy versions, output summary, structured reason codes, and protected-input references. Preserve original AI events when a human overrides them; do not copy raw resumes, transcripts, credentials, or free-text reviewer notes.

Important control gap

The access policy defines `audit.read` and `audit.write` with default deny, but the current logger writes files directly without enforcing those permissions. Add a protected audit storage boundary with access checks, encryption, durable append-only writes, and integrity verification before production use. The configured audit policy says 730 days with action `retain`; confirm its meaning before automating deletion.

Implementation order

- Confirm audit roles, trusted identities, human-review labels, and retention semantics.

- Implement structured operational logs, correlation IDs, redaction, and pipeline-stage timing/error events.

- Protect the audit store; add versioned human-review and override events with integrity checks.

- Connect metrics, calibrated alerts, and aggregate dashboard views with explicit denominators and missing-data states.

- Validate access denial, audit-write failure handling, retention/legal holds, redaction, alert routing, and dashboard roles using synthetic or de-identified records.

Supporting specifications: logging structure, metrics, alerts, dashboard, and decision-audit plan are available in `docs/day61_*.md`.